

U546 beta validation report

Report Reference number: QAD/BL/25-26/13

Date: 06.10.2025

Purpose: To validate the Beta readiness of U546 V2 battery pack (Fixed/Non-dockable battery)

Model: U546 V2

Standard Reference: TSES 779_rev06 & rev07

Background: NPD

Supplier/Change details: MIW

No of samples tested: As per below table

Part Details

SI No	Part Number	Part Name	Test reference standard	Test Conducted
1	KE243070	Battery pack assy	TSES_779	SW validation for error/warning codes, Performance and Functional tests

Test details:

SI.No	Test name	Test condition	Stage	Sample size	System requirement	Observation	Remarks
1	Capacity and energy test at different C-rates at RT	0.2C, 0.3C and 1C dch at 25degC	Beta1	1	25Ah at 0.3C, 25degC 1320Wh at 0.2C, 25degC 1300Wh at 0.3C, 25degC 1270Wh at 1C, 25degC	0.2C - 26.01Ah, 1356.4Wh 0.3C - 26.21Ah, 1361.4Wh 1C - 25.95Ah, 1297Wh	Pass
2	Capacity and energy test at different C-rates at different temperatures	0.3C dch @40degC 1C dch @40degC	Beta1	1	NA	1C - 25.93Ah, 1289.5Wh 0.3C - 26.05Ah, 1349.2Wh	Pass
3	Power and internal resistance test	Discharge at 1C for 3min	Beta1	1	NA	DCIR at 30.2% and 50.5% SoC - 65.38mΩ	Pass
4	Energy efficiency test	0.7C charge 1C discharge	Beta1	1	NA	85.70%	Observations are recorded
5	Insulation resistance test	Test voltage of 500V was applied for 60sec and resistance value was measured	Beta1	3	>10MΩ	>550MΩ	Pass

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6	Short circuit test	The terminals of the battery are shorted directly	Beta1	1	As per BMS fault matrix	Post 10 consecutive SC detection, the BMS is entering disable mode. The pack is entering active mode only after ignition reset	Pass
7	Overcharge test	As per TSES 779_rev06	Beta1	1	Cut off at Vmax>4.140V	The BMS is terminating the charge once the Vmax>4.140V	Pass
8	Over discharge test	As per TSES 779_rev06	Beta1	1	Cut off at 0% SoC	The BMS is cutting off the current when SoC reaches 0%.	Pass
9	Imbalance charge/discharge test	50mV imbalance was created in one row and pack was subjected to charge and discharge cycle.	Beta1	1	Charge: 100mV Discharge: 200mV, if previously the pack was in discharge mode	Charge: Warning at CIM>80mV, cut off at CIM≥100mV Discharge is allowed till 200mV if previously the pack was in discharge mode.	Pass
10	Passive balancing functionality test	80mV imbalance was created in one row and pack was kept in idle condition	Beta1	1	NA	The imbalance reduced from 78mV to 10mV in 7 days	Observations are recorded
11	Over temperature protection test	As per TSES 779_rev06	Beta1	1	Charge : 0 to 50degC Discharge : -10 to 55degC	Charge : Warning at 44.5degC, Cut off at 49.5degC Discharge : Warning at 50.5degC, Cut off at 55.5degC	Pass
12	Under temperature protection test	As per TSES 779_rev06	Beta1	1	Charge : 0 to 50degC Discharge : -10 to 55degC	Charge : Warning at 4.5degC, Cut off at -0.5degC Discharge : Warning at -5.5degC, Cut off at -10.5degC	Pass
13	Over current protection test	As per TSES 779_rev06	Beta1	1	Charge : > 18A Discharge : >44A	BMS is cutting off the current when it is exceeding Imax limit.	Pass
14	Regen characteristics test	As per TSES 779_rev06	Beta1	1	The pack should accept Regen current without throwing any errors.	The pack is accepting regen current as intended	Pass
15	BMS measurement accuracy	As per TSES 779_rev06	Beta1	2	Current - 0.1 to 0.4 Voltage - <0.1V	Charge: Current -0.22 to -0.02A, Voltage 0.01 to 0.135V Discharge : Current - 0.14 to -0.05A, Voltage - 0.045 to 0.015V	Pass
16	BMS node allocation test	As per TSES 779_rev06	Beta1	2	As per TSES 779_rev06	The VCU is allocating nodes as specified.	Pass
17	Battery warning and error simulation	As per fault matrix	Beta1, Beta2	4	As per fault matrix	Beta1 -Even after inducing physical damage on CFET the error was not triggered Beta2 -	Pass
18	Row to row voltage difference	As per TSES 779_rev06	Beta1	2	<50mV in loaded condition, <10mV in rest condition	Charge - 28mV, Rest - 3mV,	Pass

 Classification: **Internal**

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	measurement test					Discharge - 41mV, Rest - 6mV	
19	AV unit functionality test	As per TSES 779_rev06	Beta1	3	The AV unit should trigger at 60degC	The AV unit got triggered when BMS triggered OCT error(Tmax - 60degC). It turned off at 55degC.	Pass
20	AV unit short-circuit test	As per TSES 779_rev06	Beta1	1	As per TSES 779_rev06	The AV unit input came to normal after removing the short circuit as intended. No component failure was observed at board level.	Pass
21	Self discharge characteristics test	Case 1 : Pack to be stored at RT at 0% SoC for 14 days Case 2 : Pack to be stored at RT at shipping SoC for 60 days	Beta1	2	As per SW release note	Test case 1 : Pack to be stored at 0% SoC for 14days at RT. Pack is taking charge after 14 days. Test case2 : 0.19% per day	Pass
22	SoH loss at storage at 25degC	The pack to be stored at RT for 30 days	Beta1	1	As per TSES 779_rev06	No SoH drop was observed in 30 days	Pass
23	Thermistor pull off strength	As per TSES 779_rev06	Beta1	1	≥0.4Kgf	2kgf	Pass
24	Weld pull out strength	As per TSES 779_rev06	Beta1	1	≥4Kgf	Positive : 7.91kgf Negative : 12.72kgf	Pass
25	Vibration test	As per TSES 779_rev06	Beta1, Beta2	2	As per TSES 779_rev06	Beta1 - Post vibration test, 2 packs were subjected to HRD. 1/2 packs throwed CIM error. Weld cracks observed in multiple locations. Beta2 - The packs passed leak test post vibration test. No abnormalities were observed during functionality checks.	Pass
26	Mechanical shock test	As per TSES 779_rev06	Beta1	1	As per TSES 779_rev06	Post mechanical shock test, during HRD observed ΔT is 6°C. The board temperatures rise is within the limits. The packs passed leak test as per EOL procedure. Insulation resistance >550MΩ for both the packs. No rubbing marks were observed on the wires.	Pass
27	Battery connector durability test	As per TSES 779_rev06	Beta2	2	As per TSES 779_rev06	No damages are observed on the 52V connector in both the packs	Pass

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28	Thermal shock cycling	As per TSES 779_rev06,07	Beta1, Beta2, QP	9	As per TSES 779_rev06,07	Beta 1 - PCM leak is observed through bottom cover screws in 1/3 packs. Air bubbles were observed from AV unit cover in 1/3 packs. Beta 2 - No PCM leak was observed throughout the test. The packs have passed leak test. The observed delta T during HRD was 10degC. QP - No PCM leak was observed from the packs. During HRD, the maximum deltaT at the end of discharge was >5degC	Fail
29	Dewing	As per TSES 779_rev06	Beta1	1	As per TSES 779_rev06	The pack is communicating normally throughout the test. The pack an insulation resistance>550MΩ and has passed the leak test.	Pass
30	Label permanence test	As per TSES 779_rev06	Beta1	1	As per TSES 779_rev06	The script remained legible after the test	Pass
31	Test with loose power terminals	As per TSES 779_rev06	Beta1	1	Cut off if board temperature>90degC	OL error was triggered from BMS when positive load terminal temp reached 91degC. Cutoff in 1s. Error recovery at 72degC	Pass
32	Random vibration + IP67	Vibration as per vehicle vibration profile followed by immersion test	Beta2, QP	2	There shouldn't be any water entry	Beta2 - 1/3 packs failed leak test post vibration test. QP - No water entry was observed inside the packs	Pass
33	Continuous discharge test	Discharge at 44A	QP	2	Delta T at the end of discharge should be less than 5°C	S1 – The maximum delta T observed during the test is 7degC. Delta T at the end of discharge is 3.5degC. S2 - The maximum delta T observed during the test is 6degC. Delta T at the end of discharge is 3.5degC	Pass
34	3G vibration + IP67	3G vibration followed by immersion test	Beta2	2	There shouldn't be any water entry	No water entry was observed inside the packs	Pass



QAD - Battery Lab

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Conclusion:

The batteries have failed during the thermal shock cycling test.

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